TAQSIM NAHAWAND: A STUDY OF SIXTEEN PERFORMANCES BY JIHAD RACY

Bruno Nettl and Ronald Riddle

This study is an attempt to gain insight into an aspect of the process of improvisation in Middle Eastern music.[1] In contrast to a number of studies which generalize about this topic,[2] the purpose here is to provide concentrated documentation on the way in which a single Arabic performer improvises in one *maqam*, using one form or genre. In this respect, our study is somewhat similar to Nettl and Foltin,[3] which attempts to show the range of improvisatory techniques and of performance types used by Persian musicians when performing one portion of a *dastgāh* or mode, the *darāmad* of *chahārgāh*. It is also related to a study by Ruth Katz[4] which compares two generations of singers from a limited population group, performing one genre in an Arabic-Jewish musical tradition.

The gathering of basic musical data was carried out to some extent in the style of an experiment, and exhibits the resulting control but also the limitations of material that is especially elicited in order to provide a structured mass of data. Jihad Racy is a young Lebanese musician who was born in a village but moved to Beirut at age 18, who studied with various musicians and independently, who is proficient on several instruments, and who is well regarded by the indigneous musical community of Lebanon. He has a background in Western music as well and has studied (but not through performance) other non-European musics. He has studied ethnomusicology but has kept his performing career separate from this interest. During 1971-72 it was possible to persuade him to perform and record 100 *taqasim* for the University of Illinois Archives of Ethnomusicology. (We use the term *taqsim* for the singular, *taqasim* for plural; and *maqam* for singular, *maqamat* for plural, throughout this paper.) He was asked simply to perform and to record a *taqsim* of his choice every two or three days, and not to feel, during a performance, in any way restricted by what he had recorded previously in the series, nor before a performance to listen to the *taqasim* that he had already recorded. No audience was present. It should be pointed out that during the period in which he made the recordings, he also gave many other performances in concerts and classes. Thus, we feel that we have a sample of material whose limits are well known, and which can be regarded as thoroughly representative of its universe, i.e., the performance style of Jihad Racy, playing alone, in 1971-72.

Among the hundred *taqasim* recorded by Racy, the sixteen that are cast in *nahawand* form the basis of this study. They were transcribed, analyzed, and compared, and a part of the analytical data, as well as conclusions drawn from it, is presented here. Fourteen of the *taqasim* of the basic corpus were performed on the *buzuq*, and two, on the *nai*.

The degree to which Racy's performances are typical of Lebanese musicians in general, of musicians of his generation, of Arabic music at large, or, indeed, the degree to which they provide insight into the processes of Middle Eastern improvisation in general cannot be discussed here. Nor can the way in which his concept of *nahawand* fits into the Arabic conceptualization of that *maqam* be presently assessed. And, indeed, we cannot attempt to show the relationship of Racy's performances of *nahawand* to that of other *maqamat* in his repertory. All that we are attempting to do here is to make a case study of one musician's style of improvisation upon one model.

It may be appropriate, however, to comment upon one immediate problem, Racy's obviously unusual contact with Western music and with ethnomusicology, and the possible effects of this upon his performance. On the basis of general, careful, but unanalytical hearing of a number of Arabic musicians, we may say that despite his unusual background, Racy's performances indeed give the impression of being reasonably typical of Arabic music and thus this study may in the first instance be considered research in the field of authentic Arabic music and not an investigation of an exceptional musical phenomenon.

The *taqsim* is one of the major forms of the general tradition of Arabic music. According to Touma,[5] it is an improvised representation of a *maqam* with an organization that is rigorous in the tonal parameters and free in the temporal ones, the latter including rhythm and length. A *taqsim* may be performed between or after metric, composed pieces; it may appear alone, and, most frequently, it is an introduction to a metric, usually vocal performance. The performances in these various musical contexts may differ, and the reader must bear in mind the fact that all of the *taqasim* performed for the present study were played alone, not in the context of other music. Moreover, there exists a metric genre of *taqsim*, *taqsim 'ala al-wahda,* but it is not considered at all in this study.

Touma[6] indicates that the organization of a *taqsim* is independent of the instrument upon which it is performed. While this is true, broadly speaking, there are obvious differences between the *buzuq* and *nai* performances of Racy, and hearing recordings by other musicians has persuaded us that the instrument used in a *taqsim* performance has, indeed, a considerable bearing upon the musical style. But perhaps these differences have a greater effect in the microcosm of individual performances than upon the over-all organization.

Keeping in mind all of the limitations mentioned above, then, we have attempted, through transcription and analysis, to find out the range of performances of *taqasim* in *maqam nahawand* as performed by one musician, indicating what all or most of the performances have in common, in what respects they vary, and to identify the character of this particular concept of music in this musician's art. The analytical work proceeded in more or less conventional style, but concentrated upon the identification of structurally significant sections and their interrelationships, in so far as their motivic, tonal and temporal aspects are concerned. It is obvious that there is literally an infinite number of things that can be said about any musical composition, no matter how simple, and we have no doubt omitted many considerations with which one might well also have dealt. In the first place, we do not deal with matters of cultural, social, and aesthetic context; but that omission should not be taken as a denigration of this kind of interest. We also did not study, in detail, matters of intonation. We have approached questions of rhythm only in a very general way. And we also did not address ourselves to the origins of specific melodic or motivic units, as, for example, the degree to which some of them were learned and memorized by Racy, others are variations of known motifs, and yet others were actually improvised by him. The extent to which the material is truly "improvised" is a question we do not attempt to assess. It may be of interest to point out that Racy himself, after seeing some of the analytical data, indicated surprise at the degree to which his performances followed certain patterns.

Touma[7] comments on the question of "composition versus improvisation." The dichotomy seems to us in some respects spurious. Obviously there is no improvisatory system that does not have some canon of rules and patterns, articulated or not, as its basis. On the other hand, Touma points out, and our study corroborates, the fact that in the performance practice of all but a few undistinguished performers, no two *taqsim* performances are identical.

The Instruments and the Maqam

It seems appropriate to comment briefly on the instruments used in the performances.[8] The *buzuq* is a near-Eastern type of lute especially common in Syria and Lebanon. Related to what medieval Islamic treatises on music have referred to as *tanbūr*, the modern *buzuq* bears close resemblance to the *saz* of Asia minor and the *dotār* of central Asia. Typically plucked with a soft piece of animal horn, it has a wood sound box shaped somewhat like that of a mandolin. Its thin neck, which is about two feet long, has some twenty-seven nylon string frets tightly wrapped around it. Like the *'ūd*, the *buzuq* has a working range of about two octaves, yet, unlike this short-neck lute, it has steel strings, arranged typically in two or three courses. If two courses are used, one of them, comprising two strings, is tuned in unison to a pitch in the vicinity of middle C. The other course, comprised of two or three strings, is normally tuned either in unison or in octaves to a pitch a fourth below that of the other course. Both courses of strings are employed for performing melodies but occasionally the higher course may be used as a drone. Some *buzuqs*, including the one used by Jihad Racy, has a third course of strings. The tuning is normally Gg; Cc; cc. Yet, depending on the *maqam* to be used and the positions of the notes to be emphasized the tuning may be varied, and the absolute pitch is not relevant.

The *nai* is an end-blown cane flute most common in North Africa and the Middle East. As it has no mouthpiece, it is considered difficult to play. It has a range of about two octaves, and is furnished with six front holes and one thumb hole. The technique of playing the *nai* does not rely primarily on half-stopping or forked fingering since the holes on it are arranged, roughly speaking, in a chromatic order according to which, for example, going up a whole-step would in most cases require the opening of two holes instead of one. The fundamental register on the *nai* produces very soft tones and is usually avoided in performance, and the production of the music relies substantially on over-blowing. Since certain notes are difficult to produce on the *nai*, a performer usually uses a set of at least five instruments of different lengths, which also enables him to play conveniently at different pitch levels.

The *maqam* of *nahawand* can be defined at various levels of musical conceptualization. Most accurately, but also with greatest difficulty, it would have to be defined in terms of all of the performances, improvisations, and compositions which are regarded as being cast within it. All of the musical events from scale, ornamentation, and acceptable accidentals, to typical modulation patterns, norms of section lengths, relationship of all sorts among the sections, etc., that are considered by those who contribute to the concensus about Arabic music as appropriate, must in the end be considered as contributing to that definition. At the opposite end of the continuum of definitions, in the simplest terms, *nahawand* is a weighted scale from which tones are selected for performance and composition. In these terms, the scale used by Racy, considering G as the tonic, is G A Bb C D Eb F G, with the lower tetrachord, G A Bb C, the most characteristic configuration. Below the tonic, the F is usually altered to F#, which functions as a leading tone to the tonic.

But the first, vastly more complex sort of definition, as it relates to the performance of one individual, is really in a sense the task of this entire paper. It should be noted, however, that other points along this definitional continuum have been struck, as, for example, by Berner,[9] who indicates that certain characteristic formulae of melody and rhythm typify this *maqam*. Similarly, Touma indicates the significance of central tones and their role in delineating sections of a *maqam*.[10]

A number of other *maqamat* function as modulatory nodes. Their characteristic tetrachords or pentachords, or, where necessary, entire octave scales, according to Racy's tradition, are indicated in Example 1. Their role in the performances is discussed below. Example 1 uses G as the tonic of each *maqam*, but all of them appear at various pitch levels, both absolutely and in relationship to the tonic of the main *maqam*. Intonational details are not considered, and the intervals that are close to three-quarter tones are indicated by the symbol ᵖ , widely used in modern Persian and sometimes Arabic music to represent a "half-flat" or quarter-tone lowering.

Length of the Performances and their Sections

One of the most important ways of establishing what is characteristic of the performances as a group, and how they vary from the norm, is the study of the lengths of the performances, and of their sections, and the interrelationships of the section by length. Much of our data can best be presented in tabular form.

Table No. 1 indicates the length of each *taqsim,* the number of sections, and the length of each section. As may be seen, the lengths of the entire performances vary from about two minutes to slightly over eight minutes, with the majority (twelve) taking from just under five to just over seven minutes. The two performances on the *nai* are slightly shorter, on the average, than those of the *buzuq*.

The performances of *taqasim* everywhere in the Middle East are always divided into easily perceived sections of varying length. In our terminology a "section" indicates material that is followed by a rest of one to five seconds, and that normally ends with a distinct closing formula. The sections themselves differ enormously in length, ranging from short bits of five, eight, or ten seconds to three, four, and in one exceptional case, over six minutes. The number of sections in the *taqasim* also varies, but distinct patterning can be discerned. The smallest number is five, the largest twelve, and the majority (eleven of sixteen) are comprised of seven to eleven sections. There is some correlation between the length of a *taqsim* and its number of sections, but it is not precise and is obviously violated by several examples — e.g. No. 43, a long performance with only seven sections, or No. 69, a relatively short one with ten, and by the relationship between Nos. 1 and 91, a very short and a very long *taqsim,* each with five sections. It seems that Racy's tradition determines an approximate length of the *taqsim*, and a range of section lengths and numbers, but that these two factors are not closely interrelated.

The lengths of the sections within a *taqsim* produce definite patterning. We may divide the sections somewhat arbitrarily into four groups, according to length: minuscule (less than twenty seconds), short (twenty to thirty seconds), medium (thirty to sixty seconds) and long (a minute or more). This kind of classification reveals an interesting fact, namely, that the majority of sections in each group actually fall fairly near the center of the range of length, indicating that this grouping is not simply arbitrary but seems indeed to reflect musical functions. Using this classification, we find that of the total of 137 sections in the sixteen performances, seventy-four are minuscule, fifteen are short, twenty-six, medium, and twenty-two, long. The *taqasim* themselves then can be classified as follows in accordance with sectional arrangement:

Type A. A number of minuscule or short sections followed by a long one: five performances, typically rather short ones: Nos. 1, 32, 68, 85 and 91.

Type B. More or less regular alternation of a short or minuscule section, or of groups of such sections, with a medium or long one: Four performances: Nos. 13, 43, 51 and 69.

Type C. A series of short or minuscule sections followed by a medium or long section (or sometimes by two of these), after which a series of short sections appears, again followed by a long or medium one. This is the most common type and appears in seven performances. In Nos. 22, 44, 56 and 25, the first medium or long section appears in the second half of the performance. In Nos. 8, 39, and 55, it appears near the beginning.

These three types of *taqasim* can be considered as representing three kinds of dramatic development. Type B is essentially without climax, while Types A and C work, through the superimposition of short sections, towards climaxes that appear in the longer sections which are distinguished not merely by their length but by greater range of melodic movement and the appearances of modulations to other *maqamat*. Type C provides an intermediate and a full climax, while Type A has only the latter; and it is interesting to find that Type A, thus, is usually but not predictably represented in shorter performances.

Finally, regarding the relationships of section lengths, it is worth noting that two long sections appear together, without intervening short sections, in only three of the performances. In each case, these pairs of long sections appear at the end. Two medium length sections appear together only twice, each time before a final long section; and two short sections appear together only once. The basic structure is thus an alternation of one or more minuscule sections with a long, medium, or short one.

Minuscule phrases, on the other hand, appear in groups of two, three, or four on twenty occasions, and if one combines the minuscule and short phrases (i.e., everything under thirty seconds), one finds mainly groups of two and four, but also all numbers up to seven. Minuscule phrases appear alone in only twelve instances.

All of the *taqasim* end with long (twelve) or medium (four) sections. Eighteen of the twenty-two long sections appear in the second halves of their performances, and this is also true of fifteen out of the twenty-six medium sections. This discussion of section length gives a picture of considerable variation within a rather well-defined system of patterns.

Arrangement of Sections by Pitch

According to the literature on Arabic music,[11] the essence of musical forms associated with the *maqam* concept is their gradual ascent from low to high registers. To be sure, this type of arrangement, centering about focal points in the scale, is evident in the improvised and even the composed forms of South and West Asian music generally. The typical arrangement of a Carnatic *alapana* is to rise from material surrounding the tonic, in low register, stressing and embellishing increasingly higher tones, until a climax is reached, perhaps an octave, 10th, or 12th above the tonic, after which a quick descent closes the form. Likewise, the typical order of *gushehs* in Persian classical music is ascending.[12] An examination of Racy's performances of *nahawand* illustrates the same principle. Patterning is easily identified, but a great deal of variation is again found.

The relationship of the sections in terms of tessitura is conveniently if mechanically expressed through identification of the highest note in each phrase. A more accurate view, on the other hand, is provided by a comparison of those tones in each section which appear to have greatest structural importance, or which because of their frequency and the dependence of other tones upon them, could be regarded as temporary tonic notes dominating the individual sections. But this sort of comparison is also more subjective, and depends on a number of factors. Determination of this "tone of greatest structural importance" is a process which at least in certain cases may not be carried out identically by any two analysts. Furthermore, in some instances, a section is dominated in turn first by *nahawand* and then by another *maqam* to which the performer has temporarily modulated, or even by more than one such temporary diversion. Nevertheless, some insight may be gained by this examination.

Table 2, then, gives the sequence of highest tones and the sequence of tones of greatest structural importance for the sixteen performances of *nahawand*. Modulation is not taken into consideration, but the entire performance is treated as a unit. It will be noted immediately that the melodic movement is generally ascending, but that a zig-zag arrangement is found, particularly in the order of highest notes. The endings of the performances tend to descend somewhat, as indicated by the comparison of tonic notes. It may also be noted that early and intermediate sections often show a coincidence of highest and tonic notes, while this is not true of the final sections; this fact may be tied to the typically greater length of the final sections.

A large number of different tones appears in the table of highest notes, while the notes of structural importance are more limited and are, in the majority, confined to the tonic (or its octave), the 5th, and the 3rd. It will also be noted that in this respect the performances on the *nai* are different from those on the *buzuq,* for the former exhibit to a much smaller extent the ascending contour; the highest tones exhibiting an undulating configuration and the tonic tones appear in a more or less arc-shaped order.

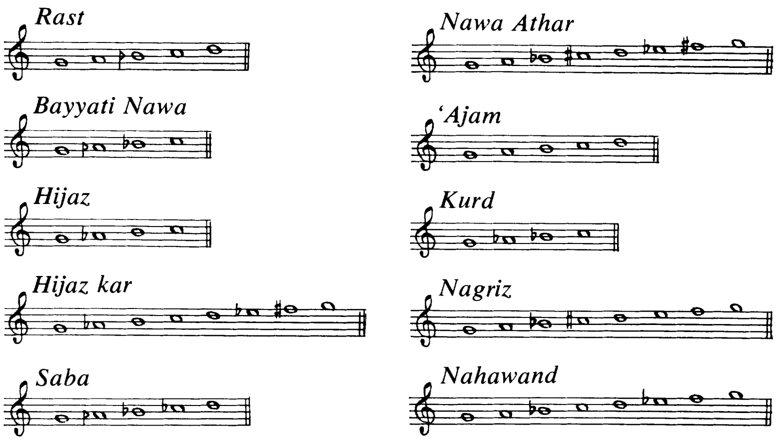
Despite the generally present tendency for the sequence of tonics to ascend, and the considerable degree of variation within this concept, it seems possible to divide the performances into types, the lines among which, to be sure, are indeed thin. Nos. 8, 22, and 39 exhibit an essentially arc-shaped contour, while Nos. 1, 25, 32, and 85 exhibit ascent with zig-zag motion. All of the rest have structurally important tones in ascending order (not counting repetitions of one note from one section to the next, and not counting the final section). The only exceptions are Nos. 44 and 56, which are ascending, excluding the final and penultimate sections. An interesting point to be made is the approximate correlation between these types and those involving the relationship of section lengths, described above. The arc-shaped order is associated with Type C, above, while all of the performances in Type B have straight ascent. Type A is associated with the zig-zag ascent group. Thus, while the two classifications do not really coincide, there is sufficient correlation to allow us to speculate on the existence of a typology of *taqsim* performances in Racy's playing.

Table 6 illustrates the way in which the higher tones are gradually introduced, from section to section, and the way in which the range of melodic material is expanded, in two *taqasim* which share the ascending form of section arrangement, and whose transcriptions are included in this paper. This table shows that the introduction of new, higher tones is indeed consistent from section to section. Typically, a tone which is introduced in one section, and appears rarely in it, is likely to be found with greater frequency in the next section which may, in turn, introduce one or two yet higher tones.

Modulation

One of the most important improvisatory techniques of *taqsim* performance involves the modulation to secondary *maqamat,* sometimes known as *taqhir*. Racy's performances make ample use of this well known phenomenon. It is our task to identify the *maqamat* to which he modulates in *nahawand* performances, their order, and the distribution of their appearance. Racy stated that two basic types of modulation are used, that in which the secondary *maqam* is built on the same tonic as the main *maqam,* and that in which it is built on another tonic, most frequently the 4th or 5th above the tonic of the main *maqam*.

Example 1, which gives the main tetrachord or pentachord of each *maqam* to which Racy modulates, indicates a total of nine such *maqamat*. Table 3 gives the order of the modulations (using abbreviations for the names of the *maqamat* and the sections wherein modulation appears).



Example 1. Characteristic tetrachords or pentachords, or entire octave scales, in the secondary *maqamat*.

Only performance No. 1 lacks modulation, and in the others, from one to five secondary *maqamat* appear. There is little correlation between length of performance and number of modulations. A quick look at Tables 1 and 3 indicates that, as one might expect, modulations almost always appear in the longer sections.

Table 3 indicates, however, a distinct patterning of modulation usage. The two most commonly used *maqamat* are *bayyati nawa* (twelve times) and *Rast* (ten). Three others appear with some frequency: *hijaz* (six), *saba* (five), and *'ajam* (four). *Nawa athar, kurd, nagriz,* and *hijaz kar* appear only once each. It is interesting to find that in the collection performed by Racy, consisting, as stated above, of one hundred *taqasim* in nine different main *maqamat,* only a few *maqamat* appear that are not used in the *nahawand* performances: *bayyati* and *siga*.

The first modulation is *rast* in eight of the *nahawand taqasim, bayyati nawa* in six, and *hijaz* in one. Those *maqamat* which appear only once come at the end of a sequence of modulatory *maqamat*. The order and frequency is similar, indeed, to practices found in the distribution of *gushehs* in Persian music, as indicated by the mentioned study of performances of the *dastgāh* of *chahārgāh*.[13] In that repertory, the first *gusheh* after the main one, *darāmad,* is almost always *zābol,* which also appears with greatest frequency, while those *gushehs* which appear rarely also are the ones that appear near the end of a performance. The conclusion regarding the improvisatory decision-making process seems obvious: a performer has in mind a typical sequence (but departs from it occasionally). At any point, however, he may decide to call a halt to the process, return to the main *maqam* (or *gusheh* in Persian music), and close the performance.

There is some association between the length of a section and the presence of modulations within it. That there is not much association between the total number of secondary *maqamat* in a *taqsim* and its overall length may thus seem curious but is indeed a fact, as indicated in Table 4. Those *taqasim* with four and five modulations are only insignificantly longer, on the average (ca. 6:30) than those with only one or two modulations (5:35).

All modulations appeared in the second halves of the performances, indicating the degree to which the main *maqam* must be established before secondary *maqamat* may appear. It is difficult to establish a typology of performances on the basis of modulation patterning. Perhaps a few *taqasim* may be singled out for arrangements that depart most from the average: a group of performances in which one or two secondary *maqamat* dominate three or four sections (Nos. 22, 39, 56, and 69); and a group in which a single long, final section experiences modulation to three, four, or five secondary *maqamat* (Nos. 32, 43, and 91). While a sample of sufficient size for statistical conclusions is hardly extant here, it is worth mentioning that there is at least a modicum of correlation with the types according to phrase length arrangement given above. The first of the patterns given in this paragraph (Nos. 22, 39, 56, and 69) agrees, with one exception, with Type C of the length classification, and the second pattern (Nos. 32, 43, and 91), with Types A and B. The fact that the majority of performances do not fall into a similar typological arrangement but, rather, occupy intermediate positions along a continuum from the one extreme type to the other indicates that perhaps true types of performances are not to be found here. Moreover, there is only the barest correlation among the three typologies attempted in this paper—according to phrase length arrangement, distribution of modulations, and over-all melodic contour—suggesting that Racy's performances of *taqsim nahawand* strike, in their configuration of parameters, at various perhaps equally spaced points of a continuum rather than representing types that can be easily distinguished. The only exception to this statement is the obviously considerable departure, from the rest, of the two *taqasim* performed on the *nai*.

Finally, regarding modulation, it may be of interest to examine the incidence of *maqam nahawand* as a secondary *maqam* among the eighty-four performances in Racy's corpus cast in a main *maqam* other than *nahawand*. Table 5 indicates the performances and the degree to which *nahawand* is present. As may readily be seen, the association of *nahawand* with *rast* and *bayyati nawa*, well established in the discussion of modulation from *nahawand* above, is established also in the counterpart. At the same time, it is interesting to see that *nahawand* also plays an important role in performances of *kurd, 'ajam, nawa athar,* and *hijaz kar,* while on the other hand, *kurd, nawa athar,* and *hijaz kar* do not play an important part in the *nahawand* performances. *Nahawand* does not appear as a secondary *maqam* in performances of *saba* and *siga,* but these are the main *maqamat* in only seven performances of the corpus.

Melodic Devices

A study of improvisation techniques was undertaken from the point of view of melodic interrelationships among units shorter than the sections, which have been our main units of consideration thus far. The rather statistical approach followed in a consideration of the interrelationships of sections could not, however, be conveniently applied, and we are forced to rely on more generalizing commentary.

In order to make it possible for the reader to find examples of the points mentioned, we present two transcriptions of *taqasim:* No. 22, performed on the *nai,* and No. 68, performed on the *buzuq*. Unfortunately, limitations of space make it impossible to publish other transcriptions of the corpus here.

Melodic movement is an area in which the group of performances is relatively unified, and in which all of the sections of a *taqsim* are also essentially alike. The movement is almost always stepwise; intervals larger than the 2nd are relatively rare except for the beginnings of sections. Such intervals are most commonly upward skips of perfect 4ths or 5ths, and they generally appear at the very beginnings of phrases and are emphasized by involving relatively long note values or notes repeated a number of times. Within this essentially scalewise melodic movement it is difficult to single out motifs that are recognizable and have a significance different from their musical environment. Exceptions are the beginnings and endings of sections. But it is also possible to characterize the melodic movement somewhat more specifically.

The favorite short melodic figure seems to be one which moves from one note to an upper or lower neighbor and back, sometimes with ornamentation. There are rarely more than three notes moving in either direction, and when this does occur, more ornamentation than normal is usually present, each note being elaborately ornamented, resulting in a sequence composed of three or four tiny bits (as, for example, in No. 22, section 4, the end, or section 6, the middle).

Repeated note figures are also pervasive, and usually occur in two forms: repeated notes of longer than average value (written here as eighth or quarter notes), which usually appear on the tonic or the fifth degree of the *maqam*; and groups of three or four short repeated notes followed by two scalewise descending notes.

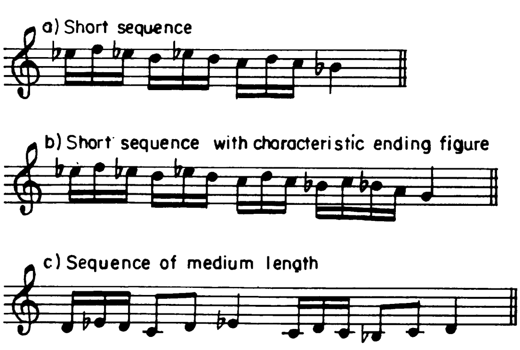
In general, the melodic material consists of short, scalewise or repeated-note figures oriented so as to emphasize the pitch that is stressed in their section. Melodies comprising motifs of extended length, i.e., more than four or five notes, are not used. Stereotyped motifs are used to begin and end sections, but the improvisatory technique in general can be characterized less as melodic improvisation than as elaborate ornamentation upon or variation of very brief motivic-rhythmic units. An exception are two *taqasim* which begin with a precomposed melody, *dūlāb,* in a version that is characteristic of *maqam nahawand,* and that is used to begin Nos. 32 and 56. *Dūlāb*[14] is a popular tune type in duple meter that is used to begin *taqasim,* but that may also be used in other musical contexts.

To the extent that one can identify significant "themes," they are concentrated at the beginnings and endings of sections. The special nature of cadential formulae has been noted as a characteristic of many musical styles. In the *taqasim nahawand* of Racy, most sections end with three notes in descending order. Interestingly, the *taqasim* performed on the *nai* differ from the rest, their sections ending more frequently with an ascending 2nd.

Many sections tend to begin with one or two motifs, one consisting of an ascending 4th followed by repetitions of the higher note, and the other comprising three notes in descending order, like the closing formula just described. These two motifs account for the beginnings of about forty percent of the sections. Some insight into the improvisation process can be gained from the observation that two consecutive sections, one short and the second long, will frequently begin with the same opening formula. The first of the pair of sections is in a sense preparatory to the second; or, the musician can be thought of as beginning a train of musical thought, finding it unproductive, and beginning the section again, with the same formula followed by what is finally more satisfactory material. Whether this process is actually operative could not be ascertained. It is possible that at one time it was, indeed, functional but that it later became a standard procedure of performance without the actual presence of this original function.

Melodic Sequence

One of the most obvious devices of melodic development is the sequence. While sequences, in the most specific form of this concept, actually occupy a small amount of the time in a performance, they provide organizing nodes of structural organization. Moreover, the same principle which generates melodic sequence can be identified at larger, less specific levels of thought.



Example 2. Sequences

A number of different kinds of sequence can be identified, and are illustrated in Example 2. It is possible to group them approximately by length, direction, degree of precision, interval, and by complexity. Most of the sequences are three-fold, i.e., a short figure is repeated twice at different pitch levels. But two-fold sequences are found, and four-fold ones as well. The number of transpositions correlates somewhat with complexity and length, i.e., longer sequences are found more frequently in two-fold form than are shorter ones, and, indeed, the occasional four-fold ones are nothing more than ornamented scales. The vast majority of sequences descend, the ascending ones being groups of two notes which are also simply elaborations of scales. The vast majority of sequences employ the interval of the 2nd for transposition; a very few examples use the 3rd.

It is possible roughly to identify short, medium, and long sequences, the short type being by far the most common. Sequences in which the transposed portions are not precise repetitions of the original portion are here referred to as anomalous sequences. Portions of melody which exhibit elements of sequential treatment, i.e., in which the same material, generally speaking, is treated at successively different pitch levels, may be referred to as sequential melodic line. If all of these types of sequential materials are considered, we find that they occupy a rather large proportion of a *taqsim*. In performance No. 55, performed on the *buzuq,* lasting six-and-a-half minutes, it is possible to identify forty-seven sequences of various sorts, twenty-one of them true or anomalous sequences, and twenty-six being sequential melodic lines. Sequential materials appear with greatest density in the last section, the longest, and they are quite rare in the short or ''minuscule'' sections, indicating that they have an important role in the more dramatic development typical of long sections and of the second halves of the performances, and also in the modulatory procedures typical of the long sections. In No. 68, lasting less than five minutes, there are twenty-seven sequences of various sorts, including seventeen of sequential melodic line.

Obviously, then, sequences and variants of the sequence concept are of great importance in Racy's improvisations. A rather large number of types of sequential treatment of motifs can be identified (as in Example 2), and their presence in the *buzuq* performances as well as those of the *nai,* and their density in the performances, also testifies to their significance. It is interesting, however, to find that sequences are not necessarily found with great frequency in all Arabic improvisations. For example, a number of *taqasim* performed on the *'Oud* in the 1930's examined by Nettl[15] indicate a large number of types of melodic movement, and sequences are, among these, much less common, though by no means absent. These recordings came from a Palestinian musician whose regional background, at least, is not very different from that of Racy, thus conceivably ruling out regional differences as the cause of the stylistic differences; generational, instrumental, and idiosyncratic styles might be relevant factors. Our task here is not, we must remind ourselves, to compare Racy's performances to those of other musicians, but rather, to try to characterize Racy's performances of *nahawand.* Comparisons with recordings made of other *maqamat,* at other times in his life, and with practices of other musicians must be made in the future and are quite likely to indicate which of the mentioned factors tend to provide unity, and which ones cause variation, within the broad stylistic framework of Arabic music.

What is more to the point, however, is some speculation about the role of sequences, for they appear with great frequency not only in Arabic but in all Middle Eastern music and, of course, elsewhere as well. We have been speaking of them in Racy's repertory as if they were to be viewed as treatment of motifs, with the shortest ones most often consisting of the more precise transposition, and with the longer ones subject to less exact reproductions at successive pitch levels. In other words, we have presented them as an important—perhaps the most important and certainly the most easily identified—method of motivic development. But it may be possible to consider them also as a function of another process in Arabic (and other Middle Eastern) music, that is, of its tendency to explore and develop successive pitch levels each centering about one main pitch. Our analysis of relationship among the sections of the *taqasim nahawand* shows that each section has a main tone, or tone of greatest structural importance, and that the relationship among the sections is in a sense defined by the relationship among those tones. It is possible to consider the sequences and the sequential melodic lines as functions of the same principle. We could state, therefore, that a basic structural principle of Racy's performances is the analogous exposition of successive, different pitch areas, and that this is done at different levels of musical thought—from the relationship of sections to each other, to the longer and not literally transposed sequential melodic lines, down to long but anomalous sequences, and further to short, literal sequences and ornamented scalar passages. This interpretation of the basic structural principles of the *taqasim* can also be used to account for the relative lack, and lack of need for, specific and distinct motifs and melodies that function as thematic germs from which further material is developed, or which clearly alternate with non-thematic, episodic material.

Other Devices

Several other devices for melodic development and continuity, less prominent than sequences, may be mentioned. One is the use of scalar passages, sometimes ornamented and therefore related to sequences, but in other cases without ornamentation. Example 3a illustrates this simple method of moving from one part of the range to another. More interesting is the variation of short motifs. Again, this can be related to the sequential treatment, since the presentation of a motif in sequence may be considered a form of variation. While we have said that it is difficult, given the non-metric rhythmic structure and the almost completely scalar melodic movement, to identify themes or motifs which could then be shown to be the subject of variation, occasionally there are identifiable bits of sufficient integrity and length to function in this role. Example 3b is an illustration. The variation of a motif usually is accomplished by going over the same ground, as it were, with some re-ordering of tones. The kind of variation of a tune that involves duplication of tones, change in tempo, and thorough, systematic departure from the norm, such as is found, for example, in Javanese gamelan music, is not involved here.

One respect in which Racy's *nai* and *buzuq* performances differ is, of course, in the wider range available on the *buzuq,* and the special use of the lower strings for unusual effects. The techniques for which the lower strings are used must be considered among the improvisatory devices used by Racy.

The lower strings most frequently have a drone function, although it is an interrupted and, indeed, interrupting drone which consists of series of repeated notes interpolated in the melody performed on the higher strings. Occasionally, the two groups of strings change roles, the lower strings being used for melodic material and the higher ones, for the drone. The drone appears most frequently on the tonic or the 5th of the scale of the main *maqam,* in about equal proportion. Other tones also appear. However, the lower string drones are much less common in the portions dominated by secondary *maqamat* than in those that are distinctly in *nahawand.* When the drone tone is something other than the tonic or the 5th, it is usually an octave reinforcement of a prominent melodic tone of the moment.

The lower strings are also used for other purposes. They echo, anticipate, or complement rhythmic patterns of the higher strings. (See the very beginning of No. 68 for an example of some of these techniques.) They are sometimes even used to play the melody, accompanied by repeated tones in the upper strings in which case the melodic material is quite contrastive to that which is typical of the style, often using large intervals—4ths and 5ths. And occasionally, the lower strings play melodic material unaccompanied by the higher strings, in which case the usual style of Racy's melody, described above, is characteristic. The contrast between low and high strings is an important component of Racy's improvisatory style and is found more or less equally in all of the *taqasim* in *nahawand.* The use of drone is particularly characteristic of the beginnings of sections, and, very generally speaking, of the first half of a *taqsim.* It thus interacts with other improvisational devices, with modulation which is more typical of later stages in a *taqsim,* and with sequence, which is also not typical of the beginnings of sections.

Ornamentation, Rhythm and Tempo

We now come to some areas of performance that are very difficult to discuss analytically and objectively. To the Western listener, Arabic music sounds extremely ornamented, but, of course, what constitutes an ornament in one culture may be part and parcel of the melodic material to another. An obvious approach would be to ask an Arabic performer to identify "ornamental tones" in juxtaposition to others, but this approach turned out to be unproductive in this study, despite the important role of ornamentation as an aspect of musical modernization and, in effect, inverse Westernization that Katz has made clear.[16]

The presence or absence of ornamentation as a concept notwithstanding, it is possible to identify something functional in the music that may best be labeled ornamentation. Let us define ornamentation as the embellishment and the resulting emphasis of a tone. We note that longer tones (eighth and quarter notes) frequently have shorter neighbors or two-tone figures attached to them. At the other extreme, we have stated that the function of an entire section may be the establishment of a structurally important tone, or of the juxtaposition of alternating tonic levels, and that we could thus regard all of the other tone material of the section as basically ornamental to that main tone. Thus, again, as in the case of melodic sequence, a basic principle of performance can be seen to operate at the microcosmic as well as the macrocosmic levels of a performance.

Racy's *taqasim* cannot be readily divided into groups according to the provenience or position of ornamentation, for in this respect his performance practice is the same in all. It is possible to see that the microcosmic ornamentation increases in the course of a performance, showing a functional relationship to the use of the drone, and to repeated note patterns that are unornamented and that are characteristic of the beginnings of sections and also of the early sections.

The rhythm of Racy's *taqasim* has not been subjected to intensive study. The absence of meter (with the exception of the initial use of the *dūlāb* melody in two *taqasim*) is part of the definition of this type of improvisation. But the absence is not complete. The grouping of tones and the use of the drone strings provide quasi-metrical organization at times, and it is further possible to identify portions of greater and lesser "metricity" within one section. A typical section is likely to alternate very rapidly between material in which there are elements of meter and those in which it is absent. Very characteristic is the kind of movement in which slow, fairly metrical material changes to quicker metrical and then quicker non-metric and, finally, slower non-metric improvisation, all within a few seconds.

It is possible to see change in degree of "metricity" from the beginning to the end of a *taqsim*. The early sections tend to be more metric, and there is movement, within a section and in the over-all performance, towards more non-metric material. Both non-metrical and relatively metrical material may be found at all stages of a performance. Relatively metrical material is associated with the use of drone strings and repeated note patterns. Thus, again, a procedure found in the microcosm of a section is also used in the over-all structure.

Another aspect of rhythm worthy of mention is the arrangement of notes of equal length into groups. Our transcriptions group the notes according to stress (each group begins with a slightly stressed note) and rests. Groupings from two to six notes of the same length are found to dominate rhythmic structure. Their frequency and their length decreases as the performance proceeds, something that is probably a function of the gradual decreasing force of metricity in the course of a *taqsim*. Many other aspects of rhythm could, of course, be examined. We are able to mention only one more: the use of stress, which is associated with volume. Alternating greater and lesser stress is a function of alternating directions of the plectrum. Beyond this, however, it is worth noting that long and repeated notes are stressed and that thus, greater volume is found near the beginnings of performances than near the end. Otherwise, while they are important components of performance, stress and volume do not seem to contribute greatly to the structure of Racy's *taqsim* performances.

Finally we shall point out that the tempo of the *taqasim* remains generally constant from beginning to end of a given piece. However, fluctuation of tempo is characteristic, taking the form, for example, of prolongation of individual tones, slight accelerations and decelerations of the tempo within phrases or larger sections, and pauses of varying duration.

The tempo accelerates slightly when tones of relatively long duration are repeated several times. (When there is a more pronounced *accelerando,* the note values are altered in the transcription, as e.g., the repeated eights near the beginning of section 5 of *taqsim* No. 68 giving way to repeated sixteenths). There is a tendency to slow the tempo slightly at cadential points of sections. Generally the concluding section—normally the longest section of the *taqsim*—is that in which the most tempo fluctuation occurs.

In their rhythmic character, the pieces for *nai* are most distinct. They reflect rhythmic characteristics that are inherent in flute performance, e.g., the capability of sustained tones and the necessity for breath-pauses. The *nai* pieces tend thus to be less metrical in character than the performances on the *buzuq*. The use of fast, rhythmically pronounced repeated-note figures, so pervasive in the *buzuq* playing, is of course not found in the *nai* improvisations.

Conclusions

The above remarks regarding Racy's performances of *nahawand* must be regarded as a pilot study. The results of the investigation tend to confirm, perhaps in somewhat greater detail but with the use of a smaller corpus, the generalizing statements of earlier publications regarding Middle Eastern improvisation. The thrust of our study has been to see whether it is possible to provide a picture of the range of performance and improvisatory practices in the experience of one musician improvising on the basis of one model, or performing what to him is basically the same material.

We find that the performances are relatively unified in terms of the use of melodic devices, ornamentation, and rhythm. In these respects, the musician does more or less the same thing in each *taqsim,* although he does not follow the same procedures in each *section* of a given *taqsim*. It is in matters of length, arrangement of sections, and modulation practice that he provides variety. Thus it is these aspects of the music which might be labeled as more truly "improvised" while those revolving about melodic devices and rhythm, which are constant throughout our sample, might best be labeled as "performance practice." While the entire performance is technically improvised, according to the usual musical definition of that term, in the sense that it is not the precise reproduction of a memorized model, we are able to hypothesize the existence of two concepts, performance practice, which involves the aspects of the music which are more or less always present in Racy's *taqsim nahawand,* and improvisation, which involves those aspects in which the *taqasim* differ from each other. Of course, the two concepts interact, particularly as both partake of certain structural principles, such as sequence and ornamentation, which have been shown to operate at the largest and smallest strata of musical thought. But it is interesting to see that those aspects of music which are usually considered by Western musicians to be the most susceptible to improvisatory variation, namely, ornamentation, rhythm, and development of motifs over short stretches of time, are the ones which contribute least to the way in which the concept of *taqsim nahawand* differs from performance to performance. And on the other hand, it is important to note the rather considerable, though obviously also strictly patterned, diversity in the larger structural aspects of the performances.

Perhaps the most significant finding is the existence of certain principles that are characteristic in the macrocosm and the microcosm of performances, and at points between these extremes. In certain parameters or elements of music, the performer carries out the same kind of musical thinking in bits of melody hardly more than a second in length, in longer segments of melody, in sections, and in entire *taqasim*. The structural integrity of these improvisations is thus considerable, and we again learn that in those cultures in which improvisation plays a major role,[17] it provides freedom for the musician to invent only within a rigorous and tightly-knit system of structural principles.

Finally, although we have presented problems of structural analysis through the intensive study of a small, rigidly controlled corpus, we should not fail to add our appreciation of the wealth of musical invention that we have encountered within the rigorously constructed network of the *taqsim* concept, and of the fine musicianship of Jihad Racy.

Notes on the Transcriptions (for Nos. 22 and 68):

It is probably impossible to render a complete, descriptive transcription in Western notation of anything, least of all of a rhythmically complex and subtle piece such as a non-metric *taqsim*. Nothing of the sort was attempted here. The transcriptions were made with the purpose of understanding the over-all structure of the performances, and the reader using them should keep this sole aim in mind.

The time values as expressed in the notes and rests of these transcriptions serve to indicate relative duration within the rhythmic flow, regardless of any slight speeding or slowing of the tempo in a given section. They are not to be regarded as metronomically exact. Neither time signatures nor bar lines have been used. Sections of the *taqasim* are demarcated by vertical dotted lines, which thus indicate pauses, generally of one to three seconds. Drone-string notes in the *buzuq* pieces are placed on a separate clef and are notated arbitrarily as whole notes, regardless of their duration. When, however, notes on lower strings have definite melodic or rhythmic patterns, they are notated accordingly.

As absolute pitch is not a concern of this study, we have for convenience notated the *buzuq* pieces as having a C tonic, the *nai* piece as cadencing on G.

Portamento is indicated by a straight line connecting note heads. Horizontal brackets above the staff enclose melodic sequences. An accidental applies only to the note that it immediately precedes.

No. 22


5.
6.
7.


8.
9.
10.


11.

No. 68


3.
4.


5.

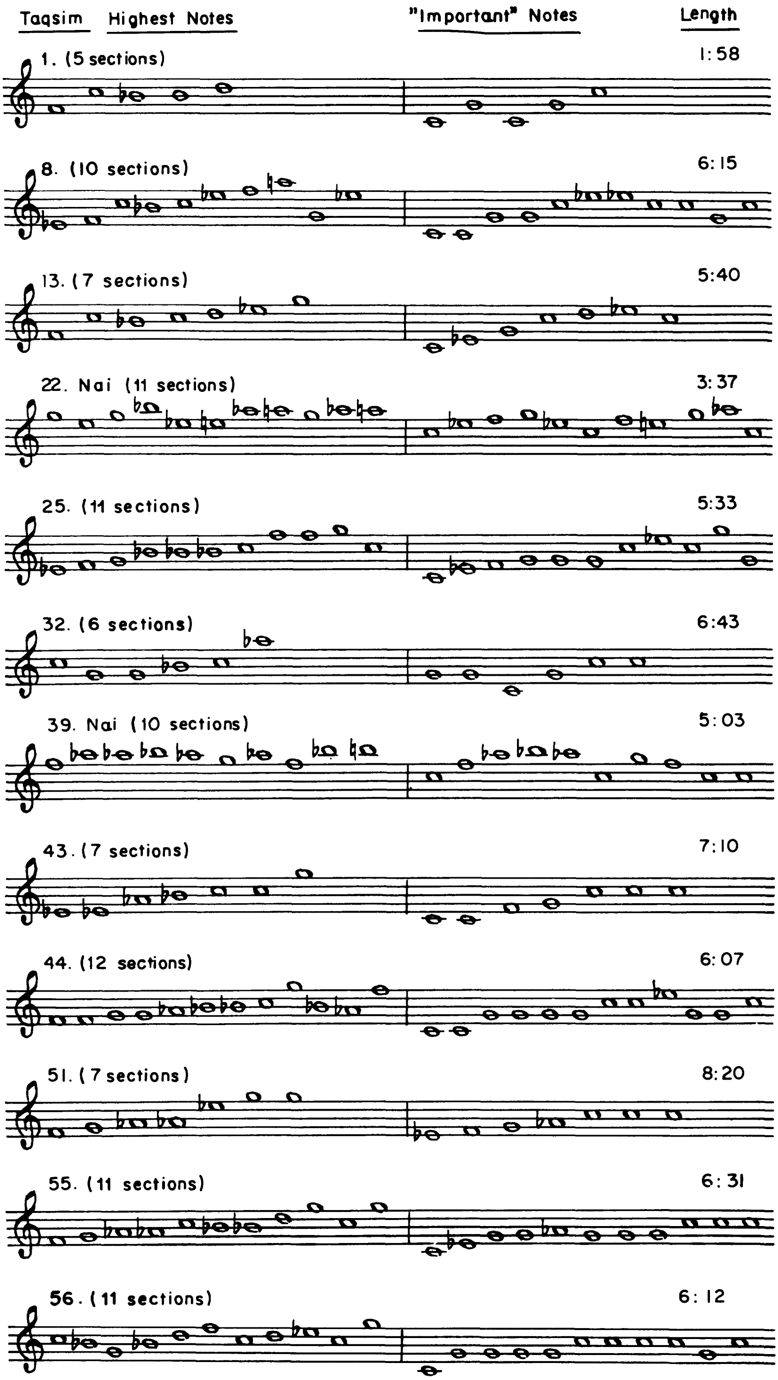

6.

Table 1

16 performances of Taqsim Nahawand, with total length, number of sections, and lengths of sections. Numbers of the performances are those of Collection No. 92, University of Illinois Archives of Ethnomusicology.

No. of Perf.	Total Length (Min. and Sec.)	No. of Sections	Lengths of Sections
1	1:58	5	:20-:13-:28-:9-:50
8	6:15	10	:14-:22-1:26-:9-:8-:13-:17-2:04-:15-1:19
13	5:40	7	:17-1:35-:15-:37-:6-:19-2:34
22 (Nai)	3:37	11	:8-:16-:5-:30-:20-:8-:47-:14-:8-:20-:37
25	5:33	11	:17-:18-:11-:19-:31-:7-:40-:20-:42-:59-1:21
32	6:43	6	:22-:18-:25-:6-1:1-3:32
39 (Nai)	5:03	10	:16-:51-:16-:42-:28-:11-:12-:4-:28-1:35
43	7:10	7	:10-:20-:47-:13-:55-:6-4:37
44	6:07	12	:13-:19-:11-:4-:31-:24-:3-:44-2:28-:4-:3-1:25
51	8:20	7	:24-:46-:10-:5-2:48-:8-3:58
55	6:31	11	:27-:6-:32-:15-:40-:8-:11-:28-1:01-:8-2:39
56	6:12	11	:20-:18-:27-:10-:42-1:11-:6-:11-:35-:40-1:22
68	4:48	6	:10-:21-:8-:47-1:05-1:56
69	5:43	10	:28-:8-:40-:10-:40-:28-:37-:6-:57-1:31
85	7:10	8	:20-:6-:12-:17-:49-:12-2:20-2:58
91	8:10	5	:19-:11-:19-:51-6:25

Table 2


Taqsim
Highest Notes
"Important" Notes
Length
1. (5 sections)
1:58
8. (10 sections)
6:15
13. (7 sections)
5:40
22. Nai (11 sections)
3:37
25. (11 sections)
5:33
32. (6 sections)
6:43
39. Nai (10 sections)
5:03
43. (7 sections)
7:10
44. (12 sections)
6:07
51. (7 sections)
8:20
55. (11 sections)
6:31
56. (11 sections)
6:12

68. (6 sections) 4:48

69. (10 sections) 5:43

85. (8 sections) 7:10

91. (5 sections) 8:10

a) Scalar passage

b) Variation

Table 3

Order of modulations and sections wherein they appear in nahawand.

A	— *'Ajam*
B	— *Bayati nawa*
H	— *Hijaz*
Hk	— *Hijaz kar*
K	— *Kurd*
NA	— *Nawa Athar*
Ng	— *Nagriz*
R	— *Rast*
S	— *Saba*

	Modulations in Order	In Sections	Length	No. of Sec.
1 -	none		1:58	5
8 -	R H S B	8, 9, 10	6:15	10
13 -	B H	6, 7	5:40	7
22 -	R	8, 9, 10	3:37	11
25 -	R B NA	10, 11	5:33	11
32 -	B S A K	6	6:43	6
39 -	R A	6, 7, 8, 9, 10	5:03	10

43 -	B H A S Ng	6	7:10	7
44 -	H R B	9, 10, 11, 12	6:07	12
51 -	B	7	8:20	7
55 -	B S A	9, 10, 11	6:31	11
56 -	R H	9, 10, 11	6:12	11
68 -	R B	6	4:48	6
69 -	R B	8, 9, 10	5:43	10
85 -	R B S H R	7, 8	7:10	8
91 -	B R Hk	5	8:10	5

Table 4

Number of secondary *maqamat* related to length of *taqasim*

No. of *maqamat*	No. of *taqasim*	Range of lengths	Approx. average length
0	1	1:58	2:00
1	2	3:37-8:20	6:00
2	5	4:48-6:12	5:25
3	3	6:07-8:10	7:00
4	3	5:33-6:43	6:10
5	2	7:10	7:10

Table 5

Nahawand as a secondary *maqam*.

Main *maqam*	Number of *taqasim*	Number containing *nahawand*
'Ajam	13	3 — 22%
Bayyati	15	2 — 13%
Bayyati nawa	8	3 — 37%
Hijaz	6	1 — 17%
Hijaz kar	12	5 — 42%
Kurd	6	3 — 50%
Nawa athar	1	1 — 100%
Rast	16	5 — 31%

Table 6

Number of notes at each pitch level, per section in performances No. 55 and 68.

																C^b Octave Higher								
	C	C#	D	Eb	EP	E	F	F#	G	Ab	AP	A	Bb	BP	B	C	C#	D	Eb	EP	E	F	F#	G
55																								
1	46		33	41			13																	
2	2		2	3			4		1															
3	24		20	36			38	1	38	13														
4	7		6	10			11		20	4														
5	30		31	41			25		27	18			2			2								
6	5		2	1			1		5	2			1											
7	2						4		24	16			13			2								
8	18		16	26		1	25	9	19	8		5	14			12		3						
9	90		49	53			19		19		13	6	15		25	87		46	49			15		1
10	9								2	2			3			9								
11	90	9	71	201	2	1	81	2	182	45	40	9	75		15	46	2	18	21			11	1	6
68																								
1	36		11	14			3																	
2	22		21	31			24		6															
3	3		2	3			6		6															
4	28		21	31		1	41	1	43	20			2											
5	71		52	65			49	6	51	37		4	28			27		6	6					
6	142		71		58		72	6	92	8	17	28	19	22	19	82		23	17					

FOOTNOTES

1. This study is part of a series of studies on improvisation in West Asian music undertaken by and under the direction of Nettl. In this case, Nettl is responsible for the basic plan of research and for writing the draft, as well as for some of the analytical work. Riddle made the transcriptions and did much of the analysis. We are indebted, of course, to Jihad Racy for making the recordings and for clarifying a number of questions. (His contribution is described in detail in the body of the paper.) We also gratefully acknowledge the help of the University of Illinois Research Board in making possible Racy's and Riddle's contributions to the project. Nettl's portion of the work was carried out while he was an Associate of the University of Illinois Center for Advanced Study. The numbering of the transcriptions and references to performances follows the numbering of Collection No. 92, University of Illinois Archives of Ethnomusicology.

2. See, for example, Habib Hassan Touma, "The *Maqam* Phenomenon: an Improvisation Technique in the Music of the Middle East," *Ethnomusicology,* 15 (1971), 38-48; *Idem., Der Maqam Bayati im arabischen Taqsim* (Berlin, 1968); Edith Gerson-Kiwi, *The Persian Doctrine of Dastga-Composition,* (Tel-Aviv, 1963); *Idem.,* "On the Technique of Arab Taqsim Composition" in *Musik als Gestalt und Erlebnis* (Vienna, 1970), pp. 66-73; Alfred Berner, *Studien zur arabischen Musik* (Leipzig, 1937); and Hormoz Farhat, *The Dastgāh Concept in Persian Classical Music* (Unpublished dissertation, UCLA, 1966).

3. Bruno Nettl, with Béla Foltin, Jr., *Darāmad of Chahārgāh, a Study in the Performance Practice of Persian Music,* (Detroit, 1972).

4. Ruth Katz, "The Singing of Baqashôt by Aleppo Jews," *Acta Musicologica,* 40 (1968), 65-85.

5. Touma, *Der Maqam Bayati. . .* pp. 18-19.

6. *Ibid.,* p. 19.

7. *Ibid.,* pp. 93-95.

8. We are indebted to Jihad Racy for notes regarding his use of the *buzuq.*

9. Berner, *Op. cit.,* pp. 68-69.

10. Touma, "The *Maqam* Phenomenon. . . ", 43. See also Gerson-Kiwi, "On The Technique of Arab Taqsim Composition," p. 68.

11. See, for example, Touma, "The *Maqam* Phenomenon," p. 41.

12. See Farhat, *Op. cit., passim;* and Nettl and Foltin, *Op. cit.,* p. 14.

13. Nettl and Foltin, *Op. cit.,* p. 46.

14. For a definition of *Dulab,* see Rodolphe d' Erlanger, *La musique arabe,* vol. 6, (Paris, 1959), pp. 180-81.

15. This study, carried out by Nettl in the middle 1950's, remains unpublished but is mentioned in his "Some Linguistic Approaches to Musical Analysis," *Journal of the International Folk Music Council,* 10 (1958), 40. In this study, it was possible to divide a *taqsim* section into segments, each dominated by a specific type of melodic contour—scalar, undulating, closing formula, etc. It seemed most convenient to analyze the structure by plotting the distribution of the various types of melodic movement, and it was possible thereby to show structural patterns that establish a norm, and that limit departures. In certain details, such as general melodic move-

ment (use of 2nds), general rhythmic character, closing and opening formulae, these *taqasim* are very much like those of Racy. In other respects they are quite different but share with Racy's performances their basic characteristics of structural patterning.

16. Katz, *Op. cit*.

17. As indicated above in footnote 15, and as is stressed in all of the publications mentioned here.